

Lab Report: EEE 4878

Group No #10

Lab No #02

LAB TITLE:

Study, Layout Design, and Verification of a Single-Stage Common Source CMOS Amplifier Using Cadence Virtuoso Layout Suite L Editor

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1 ABSTRACT

This lab report presents the design, simulation, and analysis of a single-stage common-source CMOS amplifier using Cadence Virtuoso. Three different load configurations—diode-connected, current-source, and triode-region loads—were implemented and evaluated in terms of gain, output swing, frequency response, and power consumption. Key procedures included DC operating point extraction, AC gain and phase characterization, and transient response analysis under varying bias conditions and transistor geometries. Results demonstrated trade-offs between gain and output swing across the topologies and quantified the impact of transistor sizing on amplifier metrics.

2 KEYWORDS

1. Common-Source Amplifier
2. CMOS
3. Cadence Virtuoso
4. Gain vs Swing Trade-off
5. Load Topologies

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3 INTRODUCTION

This experiment focuses on designing, simulating, and analyzing various configurations of a single-stage common-source (CS) CMOS amplifier using Cadence Virtuoso. The objective is to investigate the effects of different active load topologies on amplifier performance, including gain, output swing, bandwidth, and power consumption.

4 THEORY

The voltage gain of a common-source amplifier is

$$A_v = -g_m R_{out} \quad (1)$$

For a diode-connected load

$$A_v = -\frac{g_{m1}}{g_{m2}} \quad (2)$$

For a current-source load

$$A_v = -g_m (r_{o1} \parallel r_{o2}) \quad (3)$$

For a triode load

$$A_v = -g_m R_{on} \quad (4)$$

5 PRE-WORK DATA AND RESULTS

5.1 CS Stage with Diode-Connected Load

For a common-source stage with a diode-connected load, the voltage gain is

$$A_v = -\frac{g_{m1}}{g_{m2}} \quad (5)$$

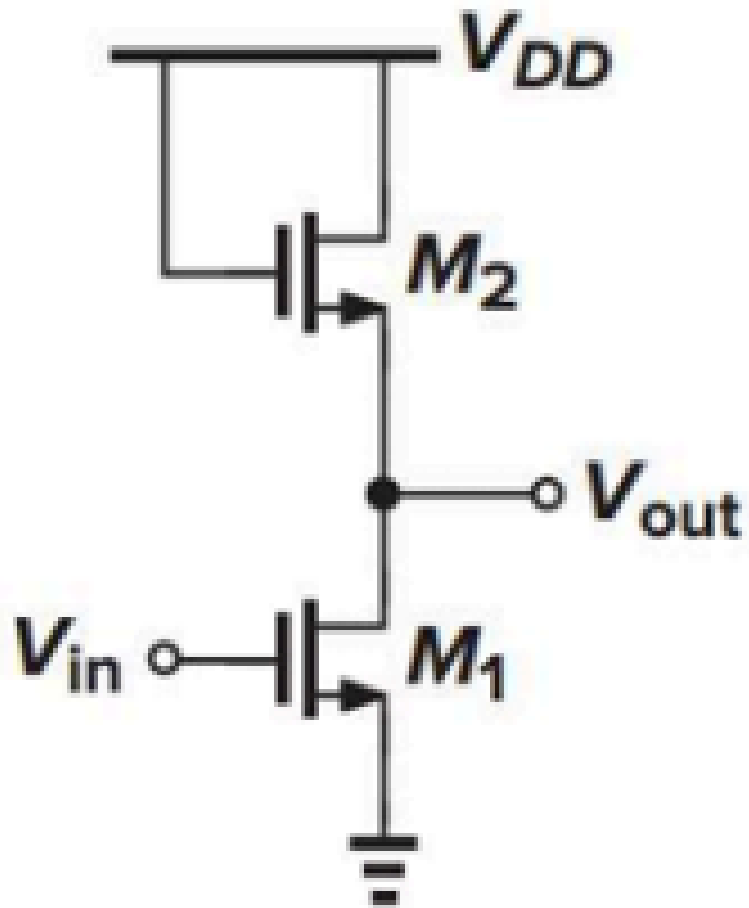


Figure 1: Diode-Connected Load Configuration

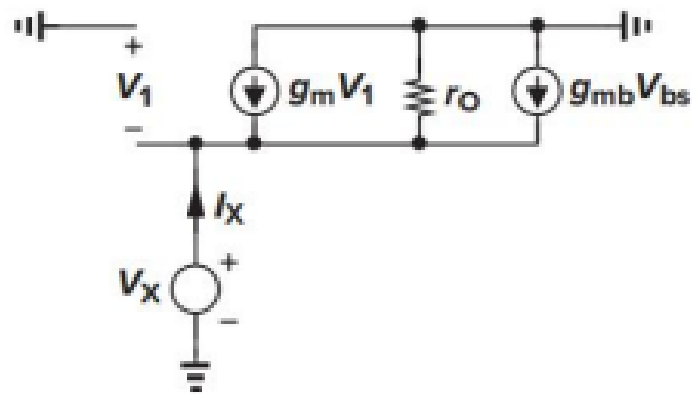


Figure 2: Equivalent Small-Signal Model

5.2 CS Stage with Current Source Load

The voltage gain of a common-source amplifier using a current-source load is

$$A_v = -g_m(r_{o1} \parallel r_{o2}) \quad (6)$$

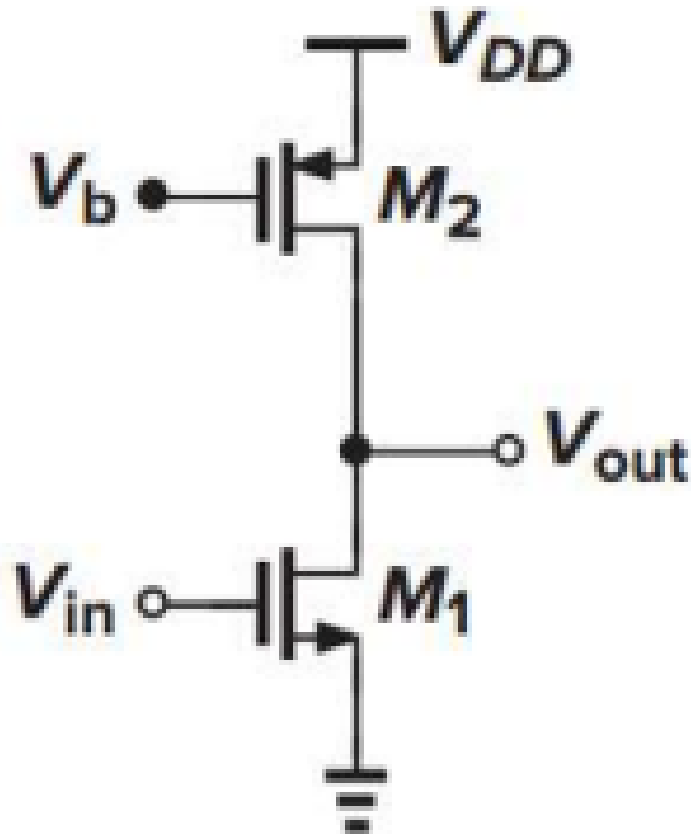


Figure 3: Current Source Load Circuit

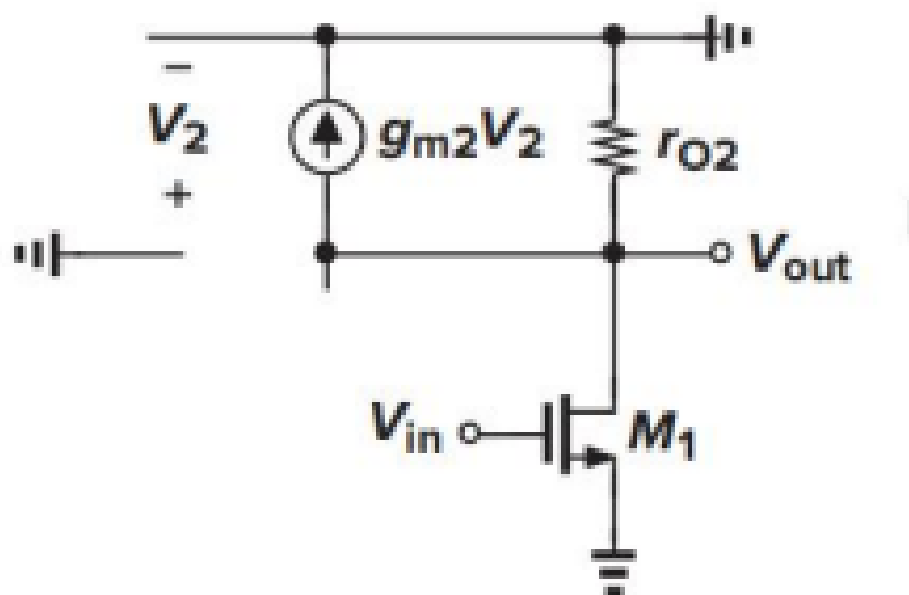


Figure 4: Small-Signal Model of Current Source Load

5.3 CS Stage with Triode Load

When the load transistor operates in the triode region, its equivalent on-resistance is

$$R_{on} = \frac{1}{\mu_n C_{ox} (W/L) (V_{GS} - V_T)} \quad (7)$$

The corresponding voltage gain is

$$A_v = -g_{m1} R_{on2} \quad (8)$$

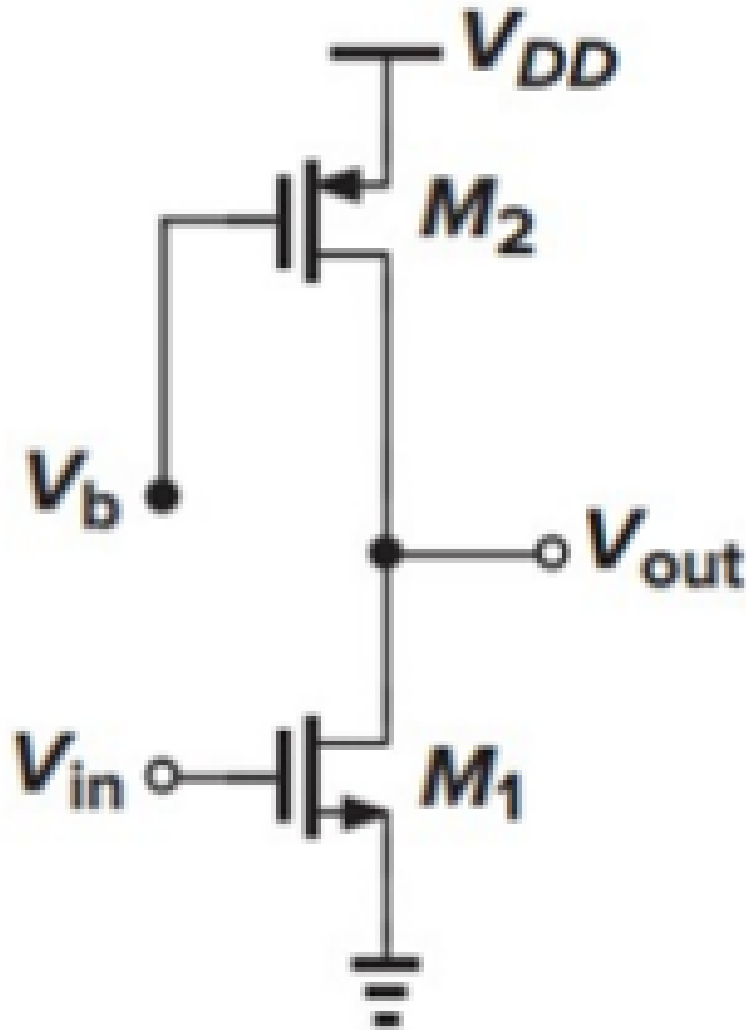


Figure 5: Triode Load Configuration

6 TOOLS USED

1. Cadence Virtuoso
2. gpd090 PDK
3. Spectre Simulator

7 PROCEDURE

1. Amplifier schematics were created in Cadence Virtuoso.
2. DC analysis was performed.
3. Transient analysis was also performed.
4. AC analysis was performed.
5. Gain and operating points were measured.
6. Different load topologies were compared.

8 RESULTS

8.1 CS Stage with Diode Connected Load

I

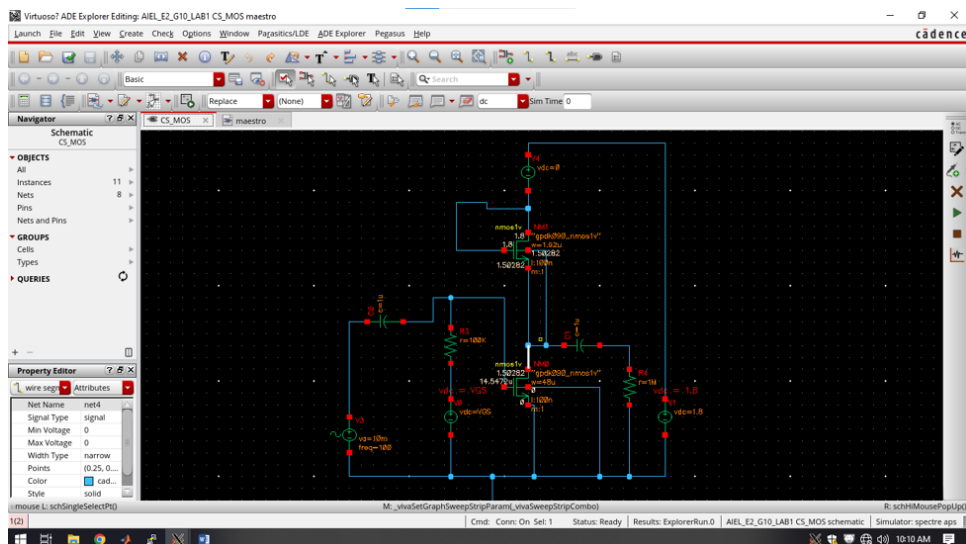


Figure 6: Diode Load Configuration

II

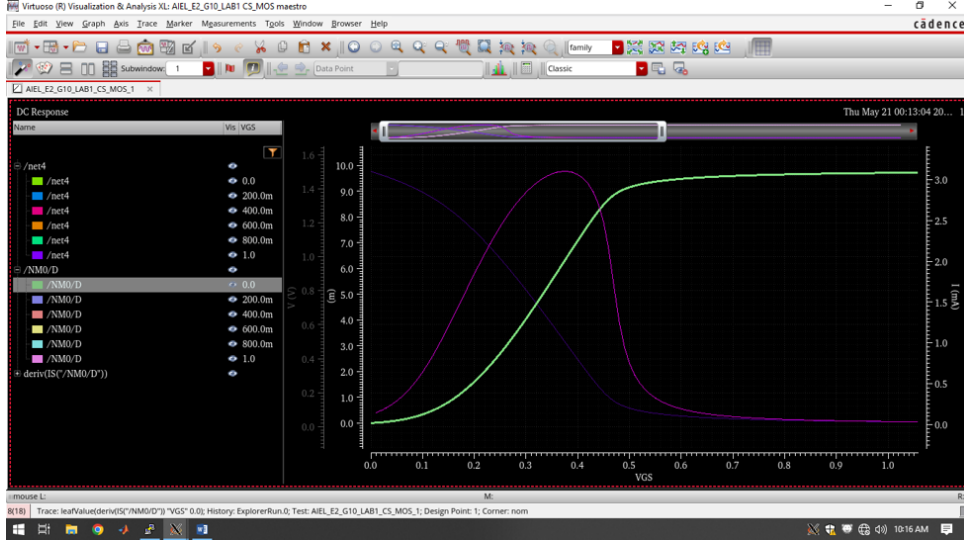


Figure 7: Drain Current vs Conductance

III

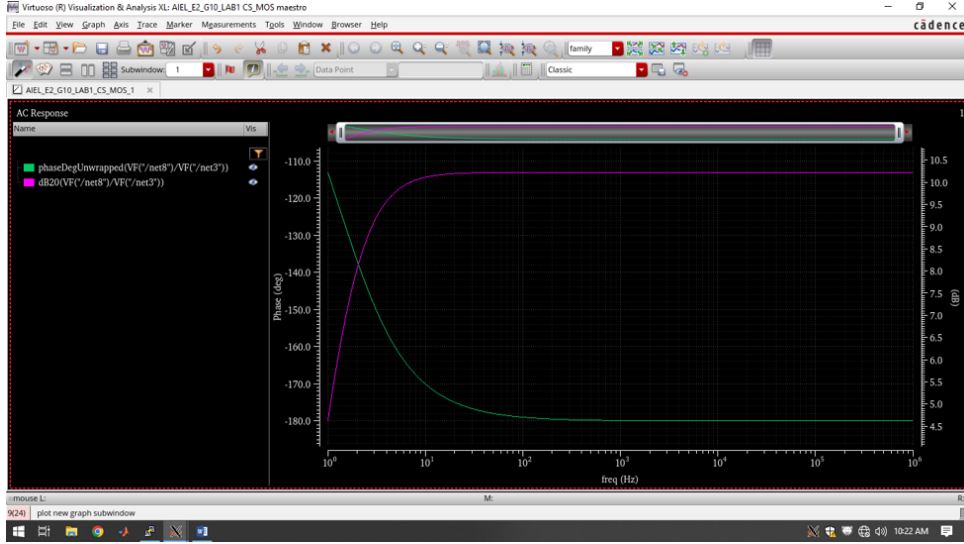


Figure 8: Gain vs Frequency with $V_{GS} = 450mV$

IV

To improve the gain of the amplifier, the transconductance of the input NMOS device can be increased by increasing its aspect ratio (W/L). At the same time, the effective load transconductance can be reduced by using a weaker PMOS diode-connected device, which increases the overall output resistance seen at the output node.

A moderate increase in bias current can further enhance g_m , although this may reduce the output resistance of the active load. These adjustments involve important trade-offs: improving g_m generally increases power consumption and parasitic effects, while reducing the load strength improves gain but can limit output swing and increase voltage headroom requirements.

8.2 CS Stage with Current Source Load

I

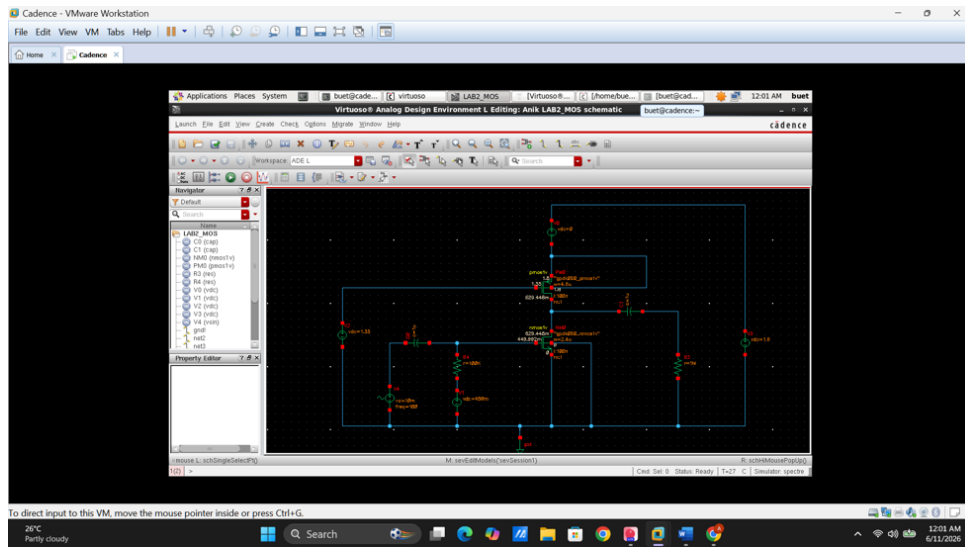


Figure 9: Schematic Diagram

II

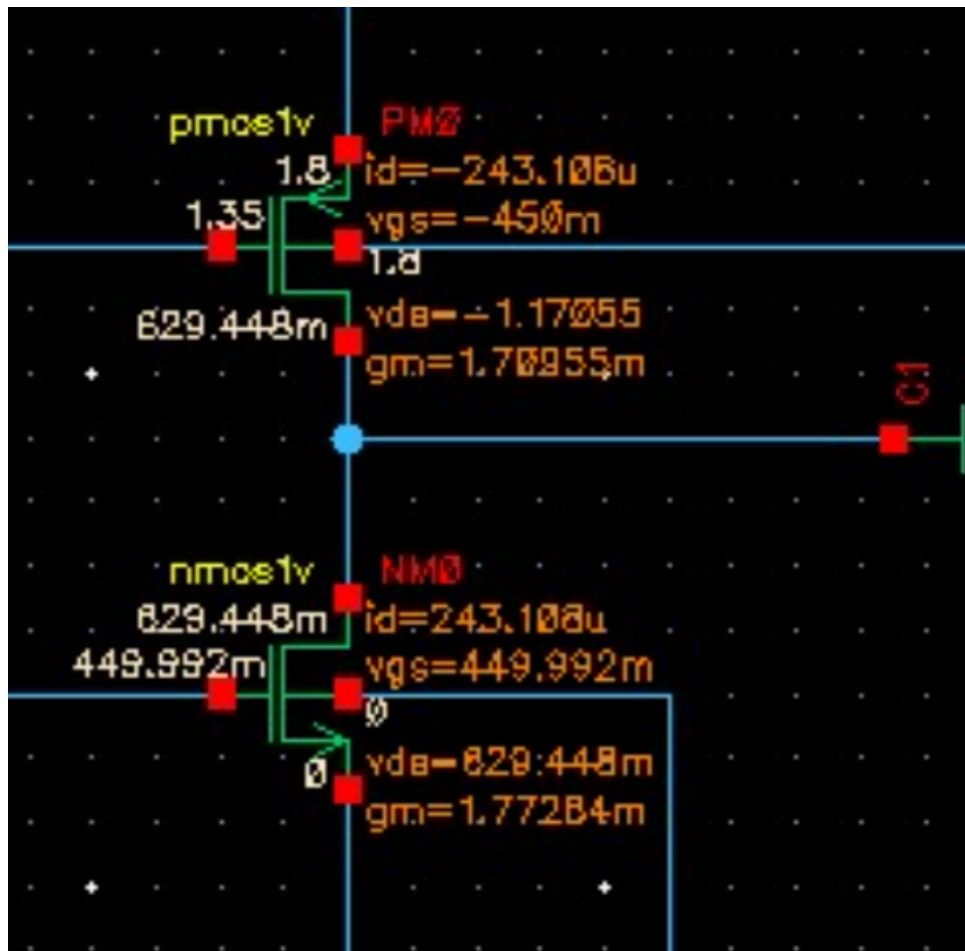


Figure 10: DC bias current $I_D = 243.106\mu A$ and $V_D = 1.71V$

III

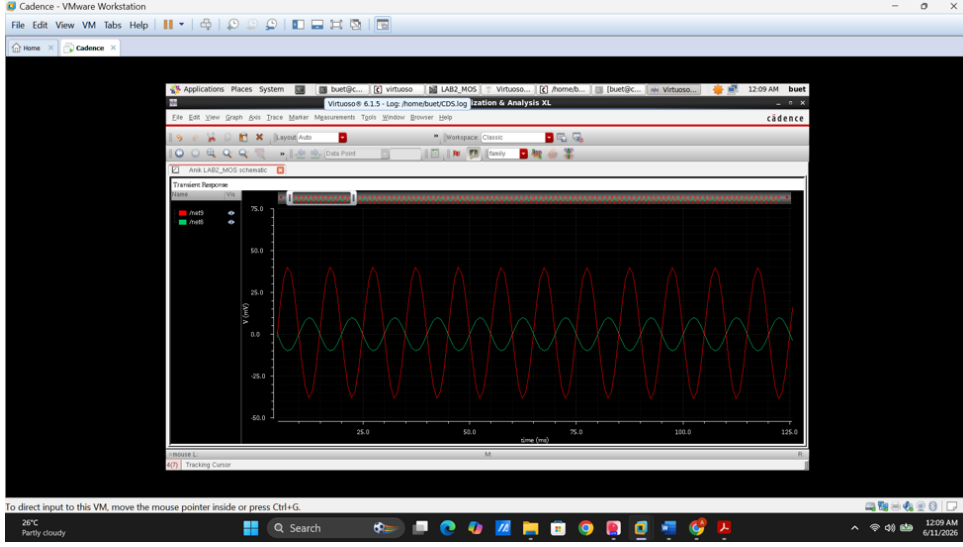


Figure 11: Small Signal Input and Output Waveform using Transient Analysis

IV Gain

$$A_v = -\frac{V_{out,pp}}{V_{in,pp}} = -\frac{80.0 \text{ mV}}{20.0 \text{ mV}} = -4.0 \quad (9)$$

V

The gain of the common-source amplifier can be improved by enhancing the transconductance of the input transistor and increasing the effective output resistance of the load network. One effective approach is to increase the g_m of the input NMOS device, which can be achieved by adjusting its aspect ratio (W/L) or operating point.

Another important method is to increase the output resistance of the current-source load. This can be done by implementing a cascode current mirror, which significantly improves the output impedance and thus increases the voltage gain.

Additionally, using longer channel lengths for both the input and load transistors helps to reduce channel-length modulation, further increasing the output resistance.

However, these improvements come with trade-offs. Increasing g_m leads to higher power consumption. Similarly, increasing output resistance through cascoding reduces the available output voltage swing, limiting the dynamic range of the amplifier.

VI

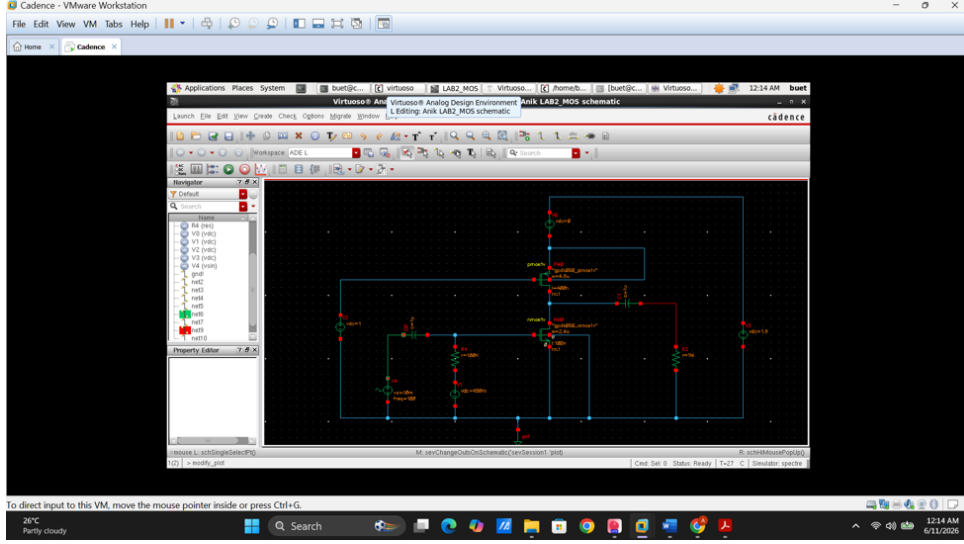


Figure 12: Schematic with modified values

VII

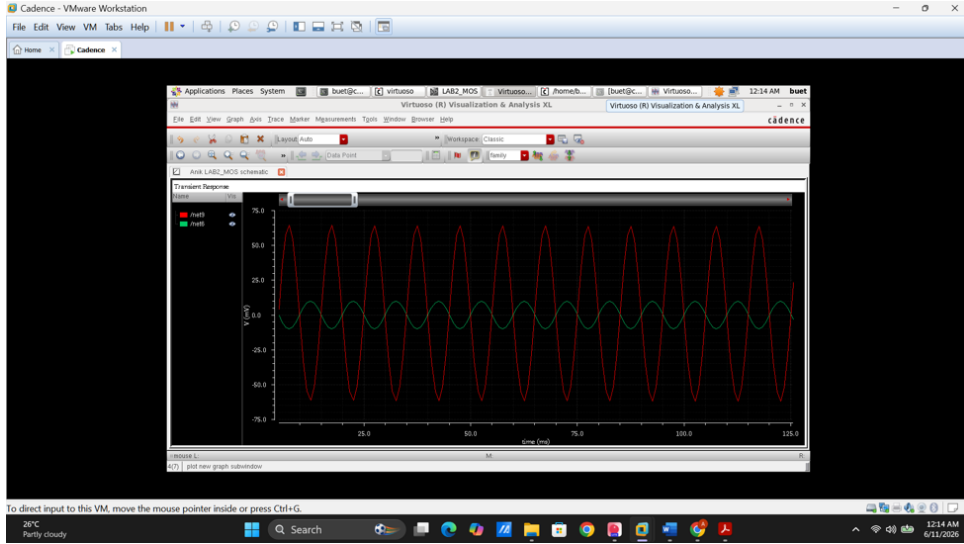


Figure 13: Small Signal Input and Output Waveform using Transient Analysis

VIII

$$A_v = -\frac{V_{out,pp}}{V_{in,pp}} = -\frac{130.0 \text{ mV}}{20.0 \text{ mV}} = -6.5 \quad (10)$$

The measured gain of the amplifier increased from 4 to 6.5 after modifying the load transistor parameters. This improvement is primarily attributed to increasing the channel length of the PMOS load transistor (M2) to 400 nm, which reduces its drain current due to the reduction in channel-length modulation effects. As the channel length increases, the output resistance of the device increases, which contributes to higher voltage gain.

Since the drain current in saturation is related to device dimensions and overdrive voltage, increasing channel length reduces current drive capability. To compensate for

this reduction and maintain proper biasing, the gate bias voltage was adjusted.

For a PMOS transistor operating in saturation, the drain current is proportional to the square of the overdrive voltage, i.e., $(V_{SG} - |V_{th}|)^2$. Therefore, reducing the gate bias to approximately 1 V increases the effective overdrive voltage, which helps restore and stabilize the drain current while maintaining improved gain performance.

8.3 CS Stage with Triode Load

I

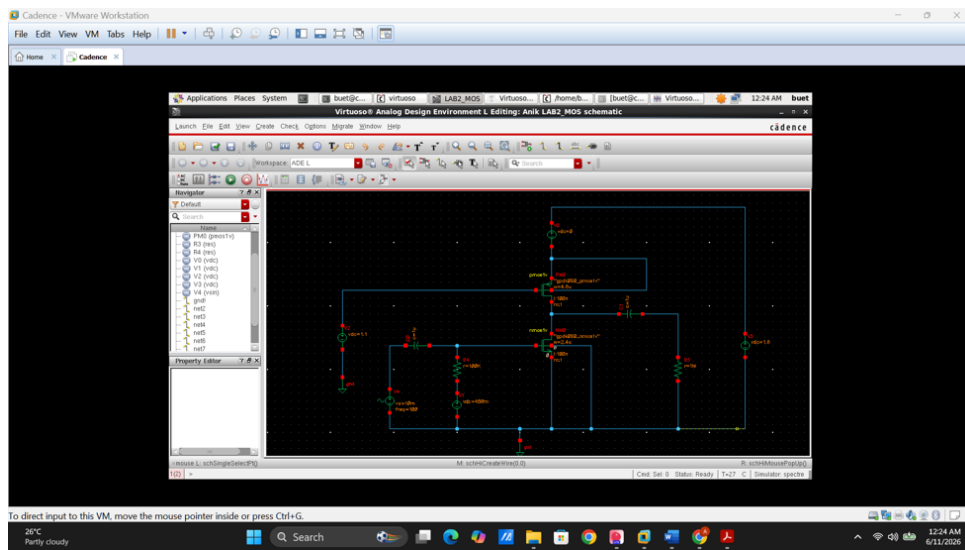


Figure 14: Schematic Diagram

II

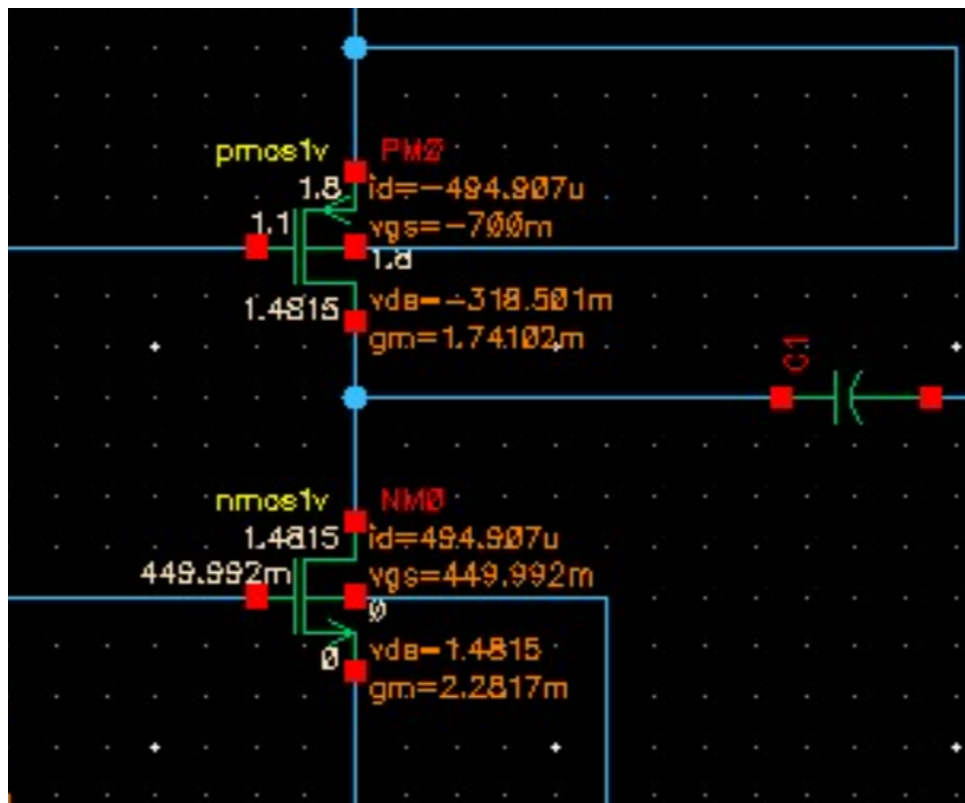


Figure 15: $I_D = 494.907\mu A$, $V_D = .7V$

III

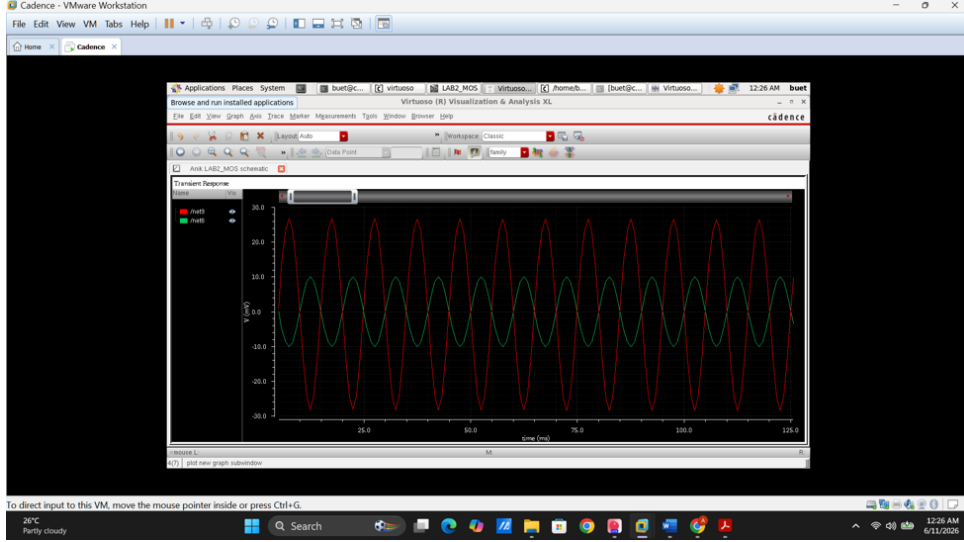


Figure 16: Small Signal Input and Output Waveform

IV

$$A_v = -\frac{V_{out,peak}}{V_{in,peak}} = -\frac{26.5 \text{ mV}}{10.0 \text{ mV}} = -2.65 \quad (11)$$

9 CONCLUSION

The experiments validated that load configuration and transistor sizing critically impact amplifier performance. Resistive loads offered simplicity but limited headroom, while diode-connected and triode loads traded gain for linearity. Current-source loads provided higher gain at the expense of reduced swing and increased circuit complexity. These results demonstrate important trade-offs encountered in practical analog CMOS amplifier design.

10 REFERENCES

1. Razavi, B., Design of Analog CMOS Integrated Circuits, 2nd Edition, McGraw-Hill, 2016.
2. Gray, Hurst, Lewis and Meyer, Analysis and Design of Analog Integrated Circuits, 5th Edition, Wiley, 2009.